EATALY

CANTEEN FOOD ORDERING SYSTEM

DESIGN THINKING PROJECT

1 **FAST**

Pre-order before the rush.

2 **SMART**

Pick up when you're ready.

3 **VISIBLE**

Know availability & demand.

TEAM

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PROBLEM DEFINITION

The project begins with a clear, actionable campus food-ordering challenge.

1 CURRENT SITUATION

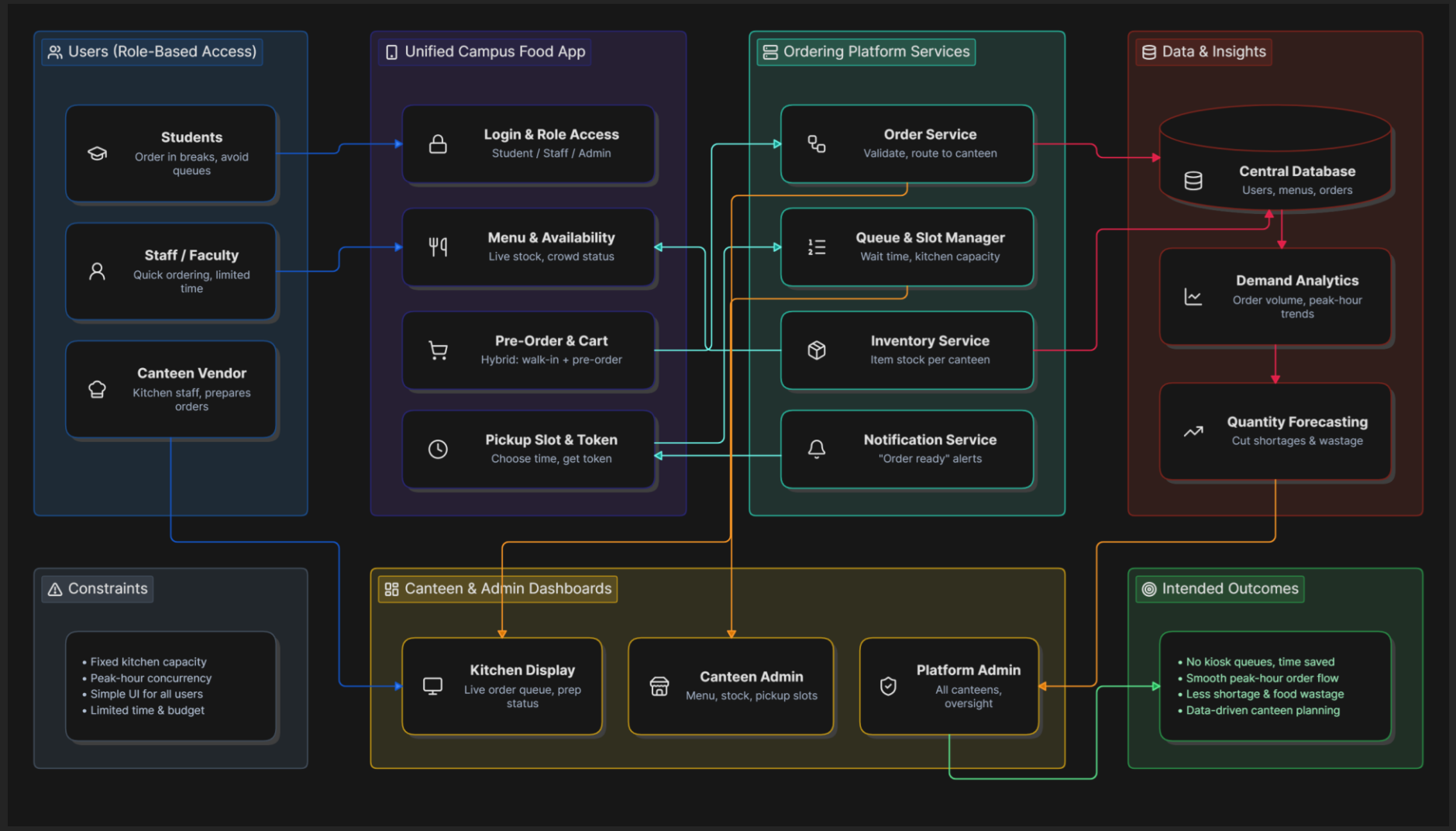
- Students face long kiosk queues during breaks.
- Food availability and crowd status are not visible remotely.
- Orders are primarily walk-in, with no scheduled pickup option.
- Canteen staff struggle to predict demand, causing shortages or wastage.

2 SYSTEM-LEVEL GAP

- There is no centralized way to track orders and demand across multiple campus canteens.
- The challenge affects students, staff, canteen teams and campus administration.
- The design must improve convenience without requiring major infrastructure expansion.

THE WHY

ARCHITECTURE DIAGRAM



OBJECTIVES

GOALS

1. Reduce or eliminate physical queues through pre-ordering.
2. Let users select a convenient pickup time.
3. Show real-time food availability across canteens.
4. Help admins track order volume and plan quantities.
5. Use role-based access for users and admins.

INTENDED OUTCOMES

TIME SAVED	Students and staff can avoid queues and collect food when ready.
ORDER FLOW	Canteens handle peak-hour demand in a more organized way.
BETTER PLANNING	Admins gain demand information for food preparation.
LESS WASTE	More accurate planning can reduce shortages and wastage.

SUCCESS = TIME + FLOW + VISIBILITY

CONSTRAINTS

The solution must work with practical infrastructure, usability and project limitations.

1 INFRASTRUCTURE

Kitchen capacity, staff and equipment cannot be scaled up quickly.

2 PEAK LOAD

The system should remain reliable with many simultaneous users.

3 USABILITY

Students, staff and admins may have different levels of technology familiarity.

4 TIME & BUDGET

A student-level project limits the features that can be built and tested.

REAL WORLD ≠ PERFECT WORLD

Where the design can evolve

Alternative directions explicitly identified in the scope sheet.

1 ORDERING MODEL

Pure pre-order
or hybrid walk-in + pre-order

2 QUEUE HANDLING

Live wait-time display
Token system
Slot-based pickup

3 PLATFORM MODEL

Single unified app
or linked canteen dashboards

4 ADMIN CAPABILITY

Basic order tracking through advanced demand forecasting

EXPLORE → VALIDATE → DECIDE

The design space stays open so alternatives can be explored and validated before final implementation.

06• EMPATHY

EMPATHY MAP

PROJECT NAME: EATALY

KRISHNAPRIYABALA J S



SAYS

- Too much time to place order
- Internet issues while paying

THINKS

- If payment is done easily, things would be better

DOES

- Waiting in crowd for around 20 mins

FEELS

- She feels hard to even place the order and recieve the invoice

EMPATHY MAP

PROJECT NAME: EATALY

HUT CAFE ADMINISTRATOR

SAYS

- Has to check each time in real time to go and check for quantity updates

THINKS

- It would be better if he had an extra manpower to handle stuff

DOES

- Sits billing continuosly even during busy hours with lot of network issues aswell

FEELS

- He feels extremely chaotic during the lunch breaks

PAIN POINTS

- Students spend a considerable amount of time waiting in queues to place and collect their food orders.
- Frequent network and payment issues make the ordering process slow and unreliable.
- The absence of a pre-order facility causes overcrowding during peak lunch hours.
- Café administrators struggle to predict daily food demand, leading to over-preparation or shortages.
- Manual order management and limited manpower make it difficult to handle large volumes of customers efficiently.
- Lack of real-time communication and inventory tracking results in delays, confusion, and food wastage.

GAINS

- Students can pre-order meals, reducing waiting time and avoiding long queues.
- A faster and more reliable ordering and payment process improves the overall dining experience.
- Better scheduling allows students to save time between classes.
- Café administrators can accurately forecast demand using pre-order data.
- Automated order and inventory management reduce workload and improve operational efficiency.
- Optimized food preparation minimizes wastage, enhances service quality, and increases customer satisfaction.

09• PROBLEM DEFINITION

PROBLEM DEFINITION

PROJECT NAME: EATALY

THE CURRENT CAMPUS CANTEEN ORDERING PROCESS IS INEFFICIENT DUE TO LONG QUEUES, NETWORK ISSUES, MANUAL ORDER MANAGEMENT, AND THE INABILITY TO PREDICT FOOD DEMAND ACCURATELY. STUDENTS SPEND EXCESSIVE TIME WAITING TO PLACE AND COLLECT ORDERS, WHILE CAFÉ ADMINISTRATORS STRUGGLE TO MANAGE PEAK-HOUR RUSH, MONITOR INVENTORY, AND ESTIMATE FOOD QUANTITIES. THIS RESULTS IN DELAYS, OVERCROWDING, FOOD WASTAGE, AND REDUCED OVERALL DINING EFFICIENCY.

FROM ADMIN

- DIFFICULTY HANDLING HIGH ORDER VOLUMES WITH LIMITED MANPOWER.
- MANUAL TRACKING OF ORDERS AND FOOD QUANTITIES.
- UNABLE TO ACCURATELY PREDICT STUDENT DEMAND.
- POOR COMMUNICATION REGARDING ORDER STATUS AND PEAK-HOUR MANAGEMENT.

FROM USER

- LONG WAITING TIME TO PLACE AND RECEIVE ORDERS.
- INTERNET/NETWORK ISSUES DURING PAYMENT AND ORDERING.
- NO OPTION TO PRE-ORDER MEALS.
- CROWDED CANTEENS DURING LUNCH HOURS.

HOW MIGHT WE

DESIGN A CAMPUS FOOD ORDERING APP THAT ALLOWS STUDENTS AND STAFF TO PRE-ORDER FOOD, SELECT A CONVENIENT PICKUP TIME, AND AVOID LONG KIOSK QUEUES WHILE HELPING CANTEEN ADMINS EFFICIENTLY MANAGE FOOD QUANTITY, ORDERS, AND DEMAND ACROSS MULTIPLE CANTEENS ON CAMPUS?

USER FLOW

1

CHECK

View food availability and current canteen conditions.

→

2

SELECT

Choose food and a convenient pickup time.

→

3

ORDER

Place the order before reaching the kiosk.

→

4

PREPARE

Canteen team receives organized demand information.

→

5

PICK UP

Collect the prepared order at the selected time.

CORE IDEA Move ordering effort away from the kiosk and make pickup more predictable for both users and canteen teams.

DESIGN OUTCOMES

DESIGN & PROTOTYPING

- Figma
- • Canva / Adobe Illustrator
- • paper / mobile prototypes

RESEARCH

- User interviews
- surveys
- literature review
- case studies

TESTING

Usability testing • feedback documentation

ACCESS

- Laptop • smartphone • internet

ITERATE → TEST → DOCUMENT → REFINE

12 • CONCLUSION

A campus food service designed around time, visibility and demand

HOW MIGHT WE...

Design a campus food ordering app that allows students and staff to pre-order food, select a convenient pickup time, and avoid long kiosk queues while helping canteen admins efficiently manage food quantity, orders, and demand across multiple canteens on campus?

EATALY

THANK YOU

TIME SAVED • ORDER FLOW • BETTER PLANNING • LESS WASTE